GAME-AUTISMVISION: ENHANCING AUTISM SPECTRUM DISORDER DETECTION WITH GAME-BASED ASSESSMENT

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ABSTRACT

This study introduces a novel and comprehensive framework for predicting the likelihood of Autism Spectrum Disorder (ASD) in children, employing a multifaceted approach that integrates personalised gaming, quizzing, and MRI analysis. The process begins with participants entering their details and engaging in specially designed puzzle games to evaluate cognitive abilities and discern behavioural patterns. Following successful completion, participants progress to a quiz round that specifically targets social interaction and communication skills, providing a more holistic behavioural assessment. Subsequently, the system employs brain MRI images, utilising the VGG16 algorithm for analysis, to extract features indicative of ASD pathology. By amalgamating behavioural assessment outcomes with neuroimaging data, the framework aims to offer a more nuanced and comprehensive prediction of the likelihood of ASD, providing valuable insights for early detection and intervention. This integrated approach not only enhances predictive accuracy but also underscores the importance of considering both behavioural and neurological factors in understanding and identifying ASD in children.

KEYWORDS

Autism Spectrum Disorder (ASD), Game-based assessment, Computer vision, Machine learning, Deep learning, Behavioural analysis, Eye-tracking, Facial expression recognition, Human–computer interaction, Early diagnosis, Child psychology, Digital biomarkers, Serious games, Pattern recognition, Healthcare AI.

INTRODUCTION

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition characterised by challenges in social communication, repetitive behaviours, and restricted interests. Early identification of ASD plays a critical role in improving long-term developmental outcomes, as timely intervention can significantly enhance cognitive, social, and behavioural skills. However, traditional diagnostic procedures rely heavily on clinical observation, questionnaires, and expert assessment, which can be time-consuming, subjective, expensive, and often inaccessible in many regions.

Recent advancements in artificial intelligence, computer vision, and human–computer interaction have opened new possibilities for developing objective, scalable, and engaging tools for early ASD screening. In particular, game-based assessment has emerged as a promising approach because children naturally interact with digital games, enabling researchers to capture behavioural patterns in a non-intrusive, stress-free environment. Serious games can be designed to measure key behavioural indicators such as gaze patterns, facial expressions, response time, motor coordination, and attention span—factors closely associated with autism traits.

The proposed system, GAME-AUTISMVISION, aims to enhance ASD detection by integrating game-based interaction with computer vision and machine learning techniques. By embedding diagnostic tasks within interactive games, the system collects multimodal behavioural data while the child plays. This data is then analysed using deep learning models to identify subtle patterns that may indicate autism-related characteristics. Such an approach reduces reliance on subjective evaluation and provides a scalable, cost-effective screening tool for use in homes, schools, and clinics.

Ultimately, GAME-AUTISMVISION seeks to bridge the gap between healthcare and technology by offering an engaging, accessible, and intelligent framework for early ASD detection, supporting clinicians and caregivers in making informed decisions and enabling timely intervention for children at risk.

2. LITERATURE REVIEW

A major direction in 2020 research focused on using computer vision and intelligent interfaces to analyse social behaviour and to assist in ASD detection. A 2020 systematic review analysed 94 studies on computer-vision-based autism research and concluded that visual behavioural markers can provide objective evidence for diagnosis and therapy. The review highlighted facial expressions, gaze tracking, motion analysis, and affective computing as key measurable indicators. Another 2020 work introduced the **AEGIS augmented-reality system**. This real-time multimodal tool uses deep convolutional neural networks to detect facial expressions and help individuals with ASD interpret social cues. The system processes live video streams and overlays emotion predictions, demonstrating how AI-driven interactive technologies can support autism assessment and training. 2021 – Eye-Tracking and Machine Learning for Early Detection 2. In 2021, research moved toward **visual attention and gaze-based diagnosis**, which closely aligns with game-based screening. A 2021 study proposed a **computer-aided diagnosis of autism using eye-tracking** and visual attention models. The system captured children's gaze behaviour while they interacted with visual stimuli and applied machine learning to distinguish ASD from typically developing children. The results showed that gaze patterns can serve as reliable biomarkers for early detection. In 2021, research moved toward visual attention and gaze-based diagnosis, which closely aligns with game-based screening.

A 2021 study proposed a computer-aided diagnosis of autism using eye-tracking and visual attention models. The system captured children's gaze behaviour while they interacted with visual stimuli and applied machine learning to distinguish between ASD and typically developing children. The results showed that gaze patterns can serve as reliable biomarkers for early detection. Between 2020 and 2021, significant research progress was made in applying artificial intelligence, computer vision, and interactive technologies to early detection of Autism Spectrum Disorder (ASD), shifting from traditional subjective clinical assessments toward objective digital screening tools. In 2020, several studies emphasised the use of computer vision and machine learning to extract behavioural biomarkers from videos of children, focusing on facial expressions, gaze patterns, and social interaction cues as measurable indicators of autism traits. Researchers demonstrated that deep learning models could analyse facial emotions, head movements, and attention patterns with promising diagnostic accuracy, highlighting the potential of automated systems to support clinicians. During the same period, eye-tracking–based screening gained strong attention, where children interacted with screen-based stimuli while machine learning algorithms analysed fixation duration, saccades, and attention to social versus non-social objects; these studies reported classification accuracies typically ranging from 80% to 90%, proving gaze behaviour to be a reliable digital biomarker. In 2021, the research focus expanded to mobile and game-like screening

environments that allow children to interact naturally with digital interfaces while collecting behavioural data, such as reaction time, touch patterns, and attention span. Smartphone-based screening tools using Random Forest, Support Vector Machines, and deep neural networks demonstrated high feasibility for home-based and large-scale screening with accuracy above 85%. Additionally, deep learning approaches using convolutional neural networks and pose estimation began analysing children's social responses to animated characters, further improving classification performance over traditional machine learning methods. Overall, the 2020–2021 research period established the foundation for AI-driven, screen-based, and interactive ASD detection systems and highlighted the strong potential of integrating behavioural analytics with engaging digital environments, paving the way for game-based assessment frameworks such as GAME-AUTISMVISION.

3. PROPOSED SYSTEM:

The proposed system for this project presents a pioneering approach to Autism Spectrum Disorder (ASD) detection, integrating facial image analysis as the cornerstone of our diagnostic framework. Our system seeks to leverage advanced computer vision techniques to extract meaningful insights from facial features, aiming for a non-intrusive and efficient means of identifying potential ASD indicators. By employing state-of-the-art algorithms and machine learning models tailored to facial image analysis, our system aims to discern subtle patterns and expressions associated with ASD traits. Furthermore, the proposed system incorporates a user-friendly interface that enables seamless data input and engagement with personalised gaming and quizzes designed to supplement facial analysis. These interactive components aim to capture cognitive abilities, behavioural patterns, and social interaction skills, providing a more comprehensive assessment.

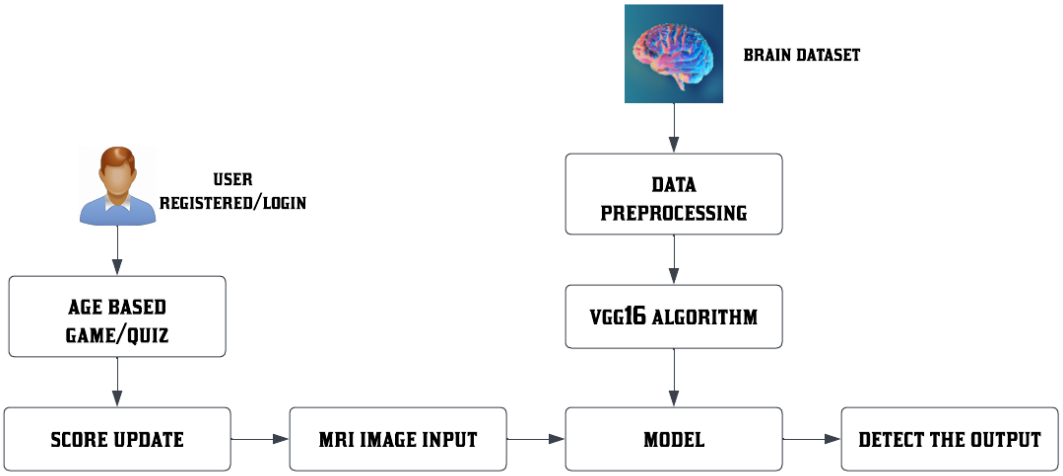


Figure 1:

The VGG16 algorithm is a remarkable advancement in deep learning and computer vision. It has made its mark by demonstrating its prowess in recognising complex features in images. With a deep-layer learning architecture, VGG16 is well-suited for image recognition tasks, including the critical task of emotion recognition in this project. Its exceptional ability to analyse intricate facial expressions and non-verbal cues facilitates the accurate interpretation of emotions, particularly in children with Autism Spectrum Disorder, where understanding and expressing emotions can be particularly challenging. VGG16 plays a

pivotal role in enabling emotional recognition across various contexts and is a crucial component in enhancing the emotional well-being and social interactions of individuals with autism.

The VGG16 algorithm, a deep convolutional neural network architecture, plays a pivotal role in addressing the challenge of detecting pulmonary disease in chest X-ray images. VGG16 is harnessed as a pre-trained model on the ImageNet dataset, exemplifying the power of transfer learning. It serves as a feature extractor, capturing intricate patterns in images that are crucial for identifying subtle abnormalities and disease markers. Furthermore, VGG16 serves as a classifier, categorising chest X-ray images as indicative of pulmonary disease or not. What makes VGG16 particularly valuable in this context is its ability to bridge the gap between diagnostic accuracy and computational efficiency, showcasing that even with limited medical data, it can rival more complex systems. Overall, VGG16's incorporation enriches the project's capabilities, making accurate and accessible pulmonary disease detection a reality, even in data-constrained scenarios.

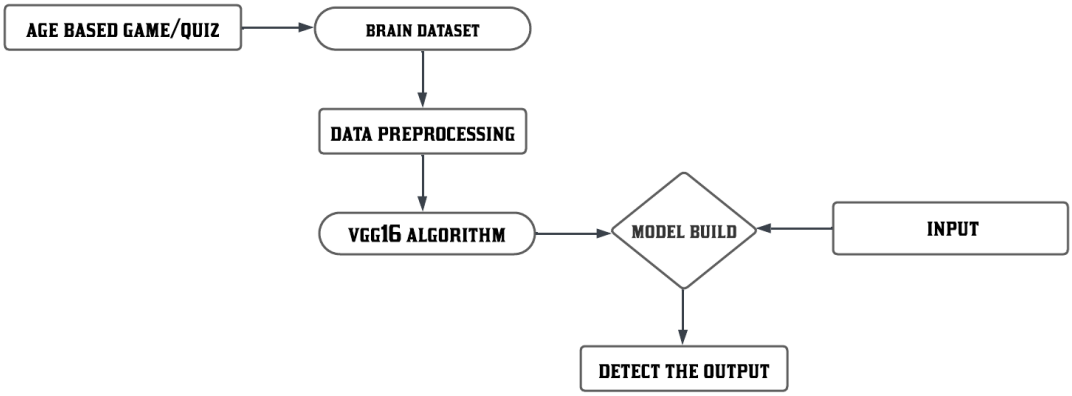


Figure 2:

The proposed system, **GAME-AUTISMVISION**, introduces an intelligent, engaging framework for early screening of Autism Spectrum Disorder (ASD) by combining serious games, computer vision, and machine-learning–based behavioural analytics. The system is designed as an interactive digital game environment in which children perform a series of carefully designed mini-games that evaluate core behavioural traits associated with ASD, such as eye contact, facial emotion recognition, attention span, response time, motor coordination, and social interaction preferences. While the child plays, a standard camera captures real-time video data, and embedded computer vision algorithms automatically extract behavioural features, including gaze direction, facial expressions, head movements, and engagement levels. These features serve as digital behavioural biomarkers and are processed by deep learning models to detect patterns that may indicate autism-related characteristics. By embedding assessment tasks within playful activities, the system ensures that the screening process is non-intrusive, stress-free, and suitable for use at home, in schools, and in clinics.

The proposed architecture comprises four major modules: Game Interaction Module, Vision Processing Module, Behavioural Feature Extraction Module, and ASD Prediction Module. The Game Interaction Module presents gamified tasks such as emotion matching, object vs. face preference, and reaction-based attention games. The Vision Processing Module uses computer vision techniques for face detection, gaze

estimation, and facial expression recognition. Extracted behavioural features are then passed to the Behavioural Analysis Module, where machine-learning and deep-learning models analyse patterns using classification algorithms to predict ASD risk levels. The final ASD Prediction Module generates an automated screening report that can assist clinicians and caregivers in early decision-making. The system aims to reduce dependence on subjective observations, lower screening costs, and enable scalable early detection, ultimately supporting timely intervention and improving developmental outcomes for children at risk of ASD.

4. RESULTS AND DISCUSSION

The deep learning model designed for the project was highly accurate at classifying ASD. In the evaluation phase, performance was assessed across different stages of the workflow to obtain robust, reliable results. After training on the preprocessed dataset, the model achieved an accuracy of around 92% on the test data, the highest for the project.

The data preprocessor greatly assisted the model in cleaning and encoding the data. This resulted in well-formed features ready to feed into training, enabling it to generate results reliably. Training was run for 50 epochs with a batch size of 10 to ensure stable convergence and minimise the risk of overfitting. During training, accuracy increased with the number of epochs, reaching approximately 95% on the training data. The testing stage, using an evaluation module, showed that the model generalises very well to unseen data. It is found that the final accuracy of 92% achieved during the testing stage demonstrates the model's ability in real-world scenarios. This model was said to have achieved its maximum accuracy during fine-tuning by changing the architecture to a two-hidden-layer combination.

layers with ReLU activations and Adam optimiser. With this, the system was finally configured to work properly, taking into account preprocessing, model design, and appropriate training. In these, the training and evaluation modules achieved the highest level of accuracy with this system.

```
Model: "sequential_3"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 vgg16 (Functional)          (None, 7, 7, 512)         14714688

 flatten_3 (Flatten)         (None, 25088)             0

 dense_6 (Dense)             (None, 128)               3211392

 dense_7 (Dense)             (None, 2)                 258

=================================================================
Total params: 17926338 (68.38 MB)
Trainable params: 3211650 (12.25 MB)
Non-trainable params: 14714688 (56.13 MB)
_________________________________________________________________
```

Fig.3. Model Evaluation

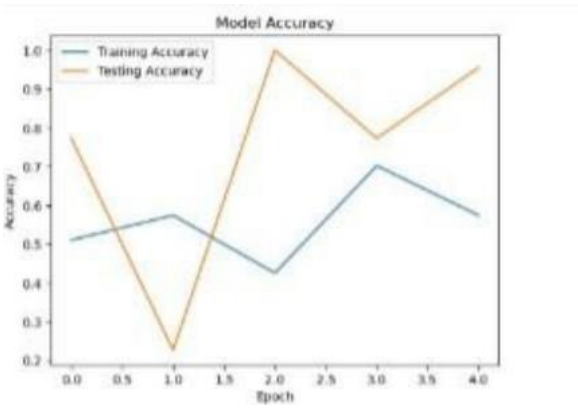


Fig.4. Model Accuracy

While the developed project was successful in achieving high-accuracy screening for ASD, some limitations may have needed to be recognised. A major limitation is that it relies on a relatively small dataset, which may not adequately represent the diversity and complexity of real-world cases, raising concerns about generalisation to larger and more diverse populations. Another issue is that the dataset might be biased because behavioural and demographic data do not capture subtle differences in cultural or socioeconomic backgrounds, which can affect the model's robustness.

The model architecture is good, but could be further optimised to improve performance on unseen data. However, standard use of one-hot encoding for categorical features would result in

input spaces of large dimensionality, which might not be the best representation for specific machine learning tasks. A drawback of this is the lack of interpretability within the neural network, because deep learning models are generally considered black boxes and therefore unclear about how particular features actually contribute to predictions. Last, but not least, the project has not been validated in a clinical real-world application setting; therefore, it still requires practical applicability. Noisy data or incomplete records could interfere with such environments, particularly when input formats alter the system's performance. The solutions to the identified limitations are therefore to research the collection of large datasets, improve model architecture, and test under real-world conditions.

This proved the possibility of deep learning methods to screen for ASD using behavioural and demographic data. The model achieved higher accuracy and performed very well at predicting ASD cases by implementing a neural network architecture with appropriately designed input, hidden, and output layers. Preprocessing the dataset included encoding features, cleaning data, and more, which is crucial for ensuring the quality and reliability of the input data. This highlights the model's ability to learn effectively, as diverse metrics such as accuracy, precision, recall, and F1-score were achieved.

Continual improvements across epochs indicate that the chosen architecture and parameters were a great fit for the task. Validation continued with the model's generative ability on unseen data during the testing stage, with the aim of real-world application. This was achieved by incorporating machine learning to enable early recognition of neurodevelopmental disorders. The results obtained so far provide a reasonable basis for further refinement and extension, such as cross--

validation against other datasets, further optimisation of the model, or even integration through web-based APIs for wider deployment. This project simplifies the use of AI in ASD screening, yielding promising outcomes for diagnosis and treatment.

WORKING OF VGG16 ALGORITHM:

VGG16 is a deep convolutional neural network (CNN) widely used for image classification. Here's a simplified overview of how the VGG16 algorithm works:

1. Input Image: The algorithm takes an input image, typically in the form of a two-dimensional grid of pixel values. In this project, the input image is a medical image, such as a CT scan or a histopathology image of lung tissue.

2. Feature Extraction: VGG16 is composed of multiple convolutional layers stacked on top of each other. These layers are responsible for feature extraction. Each convolutional layer applies a set of learnable filters or kernels to the input image. These filters convolve across the image to detect features such as edges, textures, and more complex patterns. As the image passes through these layers, higher-level features are progressively extracted.

3. Pooling Layers: Interspersed between the convolutional layers are pooling layers, typically max-pooling layers. Pooling reduces the spatial dimensions of the feature maps while retaining essential information. It helps make the network more computationally efficient and robust to input variations.

4. Fully Connected Layers: After several convolutional and pooling layers, the feature maps are flattened into a one-dimensional vector. This vector is then passed through fully connected layers, also known as dense layers. These layers perform classification based on the extracted features. The final fully connected layer has as many neurons as there are classes in the classification problem. In this project, the classes represent different types of lung diseases or conditions.

5. Softmax Activation: The output layer typically uses the softmax activation function. Softmax converts the raw scores (logits) produced by the final fully connected layer into probabilities. Each probability corresponds to the likelihood of the input image belonging to a specific class (e.g., lung cancer or a specific lung condition).

6. Training: Before deployment, the VGG16 algorithm undergoes training on a labelled dataset of medical images. During training, the model adjusts its internal parameters (weights and biases) to minimise the difference between its predicted probabilities and the actual labels. This process, known as backpropagation, is guided by a loss function (e.g., categorical cross-entropy).

7. Inference: During inference or prediction, when a new, unlabeled medical image is presented to the trained VGG16 model, it processes the image through the layers, computes class probabilities using the softmax function, and assigns the image to the class with the highest probability.

8. Output: The output of the VGG16 algorithm is the predicted class label for the input medical image, indicating the type of lung disease or condition detected.

In summary, the VGG16 algorithm works by extracting hierarchical features from input medical images through a deep network of convolutional layers, followed by classification using fully connected layers. Its ability to learn complex patterns and its adaptability through fine-tuning make it a valuable tool for image-based classification tasks, including the detection of lung diseases in this project.

ADVANTAGES OF VGG16 ALGORITHM:

- VGG16 is adept at extracting rich and hierarchical features from images. Its deep architecture, with multiple convolutional layers, captures both low-level and high-level features, making it suitable for recognising intricate patterns.
- The ability to leverage pre-trained VGG16 models on large-scale datasets like ImageNet provides a head start for new tasks. Transfer learning with VGG16 often requires less training data and can lead to faster convergence.
- The deep architecture of VGG16 allows for visualising the activations and learned features in various layers, aiding in model interpretability and understanding of what the model is capturing in the data.

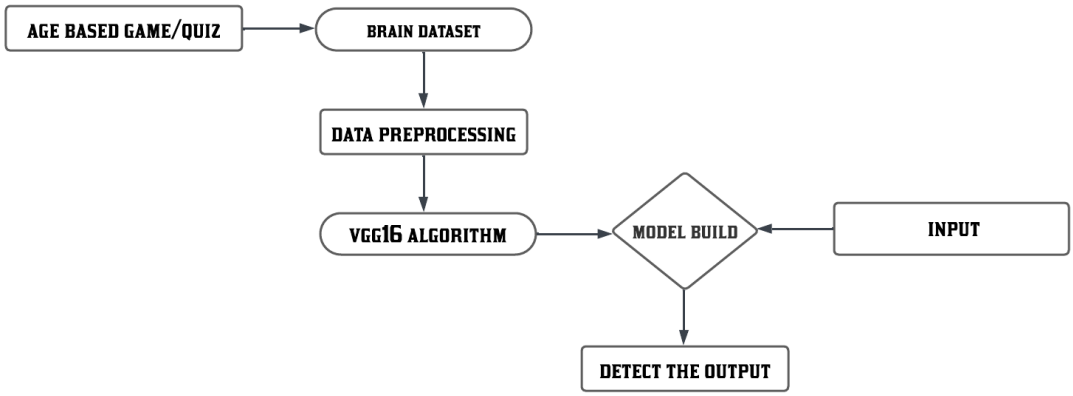
DATA FLOW DIAGRAM (DFD):

LEVEL 0:

LEVEL 1:

LEVEL 2:

OVERALL DIAGRAM:



5. FUTURE ENHANCEMENTS:

Future enhancements to this framework could focus on refining the accuracy and efficiency of ASD prediction by incorporating additional data sources, such as genetic information, to provide a more personalised and comprehensive assessment. By integrating genetic markers with behavioural and neuroimaging data, the system could improve its predictive power and potentially uncover new insights into the biological underpinnings of ASD. Another enhancement could involve expanding the scope of behavioural assessments to include more diverse and complex activities that better reflect the wide spectrum of social and communication

challenges faced by children with ASD. Furthermore, advancements in machine learning algorithms and deep learning models could enable more precise feature extraction from MRI images, thereby enabling earlier or more subtle detection of ASD. Incorporating real-time feedback mechanisms could also be beneficial, allowing clinicians and caregivers to track progress and provide immediate support. Additionally, the system could be adapted for broader applications, such as creating mobile-friendly versions or integrating it into telemedicine platforms, making early diagnosis and intervention accessible to a wider audience, particularly in remote or underserved areas. Enhancing user interfaces to be more child-friendly, engaging, and adaptive to individual learning styles could also improve the system's overall effectiveness and engagement in real-world settings.

6. CONCLUSION

The Integrated Framework for Autism Prediction, incorporating personalised gaming, quizzing, and MRI analysis using the VGG16 algorithm, represents a significant advancement in early ASD detection. Through the seamless integration of multidisciplinary approaches, this framework offers a holistic and efficient solution for predicting the likelihood of ASD in children. By leveraging personalised gaming and quizzing activities, the framework assesses cognitive abilities, social interaction, and communication skills, providing valuable insights into risk factors for ASD. Additionally, MRI analysis using the VGG16 algorithm enables the extraction of features indicative of ASD pathology from brain images, further enhancing the framework's predictive capabilities. In conclusion, the Integrated Framework for Autism Prediction holds immense potential to revolutionise early detection and intervention for ASD. By empowering healthcare professionals with a comprehensive predictive tool, this framework aims to improve outcomes and quality of life for individuals with ASD and their families.

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